



FINAL PROJECT REPORT

# EQUINOX 101

Project Helios to Andromeda

Solar Flare Mechanics – Space Weather Forecasting – Andromeda Astrophotography

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## 1 Theoretical Module: Solar Flare Mechanics

### 1.1 Task 1 - The Sun's Explosive Energy: Solar Flares

(a) *Explain in your own words what it means for magnetic field lines to “store energy”. Use a simple analogy from everyday life - for example, a stretched spring or a bent ruler - to describe how energy can be stored in a physical system and suddenly released.*

Similar to how a mousetrap stores potential energy in its spring when wound, or a bow stores energy in its string when an arrow is nocked, the magnetic field lines in the sun which store electromagnetic energy. When it reaches a breaking point it “snaps” - similar to how a mousetrap springs back from a higher tension to lower tension position - and magnetic reconnection occurs, due to which the field rearranges into a lower energy configuration releasing all the excess stored energy.



Figure 1: Tom discovers that releasing stored potential energy can be a rather energetic experience. Magnetic reconnection follows the same general idea except on a much larger scale instead of a sore finger.

(b) *When a solar flare occurs, the released energy goes into three different things: heating the surrounding gas to temperatures of tens of millions of degrees Celsius; accelerating charged particles (electrons and protons) to near the speed of light; and flinging blobs of gas outward through space. In your own words, briefly describe what you understand about each of these three effects. You do not need to use any equations.*

When a solar flare occurs a vast amount of energy is suddenly released. Some part of this energy goes to heating the surrounding gas in the Corona to millions of degrees in temperature, making it much hotter than normal. Another part accelerates charged particles such as electrons and protons to near the lightspeed allowing them to travel rapidly through space. Finally Some of the energy ejects the surrounding ionised gas into space, increasing its kinetic energy in an event

known as *Coronal Mass Ejection* (CME). These eruptions can travel through the solar system and cause geomagnetic storms on earth, stripping away its magnetic field and disrupting our satellites and communication systems.

*(c) Satellite communications, GPS navigation, and power grids on Earth can all be disrupted during a large solar flare. Suggest one reason, in physical terms, why a burst of high-energy particles from the Sun might interfere with electronic systems on Earth or in orbit. (A one or two sentence explanation is sufficient.)*

The high energy particles damage sensitive equipment in space and due to their electromagnetic fields, disturb the ionosphere which disrupts radio communication and GPS networks.

## 1.2 Task 2 - Magnetic Fields and the Stability of Solar Structures

*(a) Think about winding up a rubber band around your fingers. Describe, in everyday language, what happens physically as you twist it more and more, and why it eventually snaps or flies off. How is this a useful analogy for a twisted magnetic loop on the Sun?*

It is very similar analogy as the potential energy stored in the rubber band increases as you twist and at last reaches its breaking point when it suddenly snaps, similar to how twisting of the magnetic field line loops increases its energy until it reaches its breaking point and it releases all the excess energy in an instant.

*(b) Imagine holding a beach ball underwater. As long as the pressure of the water is strong enough, the ball stays submerged. If you move the ball to shallower water (where the pressure is lower), it becomes harder to hold down. Explain how this is a useful analogy for the torus instability of a solar magnetic loop. In your answer, identify what the “beach ball”, “the water pressure” and the “shallower water” each represent in the solar context.*

This analogy is pretty useful as it describes the torus instability in a very intuitive way. In a torus instability the magnetic field loops and the underlying gas which follows along the paths of the loops, is similar to the “beach ball”. Here the “water pressure” refers to the overlying magnetic field which holds the loop down. “shallower waters” mean the weakening of magnetic field “water pressure” with height, therefore making it harder to hold the gas loops together until they break free from the magnetic field “pressure”.

*(c) Why might it matter whether a magnetic loop erupts or stays confined? Give one example of a consequence that an eruption could have outside the Sun. A brief, qualitative answer is expected.*

It matters as if a magnetic loops erupts, it releases a huge wave of radiation and high-energy particles towards earth which can disrupt our magnetosphere and ionosphere.

### 1.3 Task 3 - Why Surface Measurements Tell Us About the Corona

(a) *Think about the skipping rope analogy. If one person holding the rope starts moving their hands in circles, what happens to the rope between them? How does this describe the way that motions at the solar surface can inject energy into the overlying corona?*

Moving one end of a skipping rope twists and tangles the rope, storing energy in it. Likewise, the churning motions at the Sun's surface twist the footpoints of magnetic loops anchored there, increasing the magnetic energy stored in the corona until the field becomes unstable and releases that energy in a flare.

(b) *In physics, pressure tells us which “side” dominates an interaction. The plasma beta ( $\beta$ ) is defined as the ratio of gas pressure to magnetic pressure:*

$$\beta = \frac{P_{\text{gas}}}{P_{\text{mag}}}.$$

*In your own words, explain what a value of  $\beta \ll 1$  implies about whether the gas or the magnetic field is in control. Then explain what  $\beta \gg 1$  implies. Where on the Sun is each condition found?*

The plasma beta,  $\beta$ , is the ratio of the gas pressure to the magnetic pressure. If  $\beta \ll 1$ , the magnetic pressure is much larger than the gas pressure, so the magnetic field dominates the motion of the plasma. The charged particles are forced to move mainly along the magnetic field lines, and the field controls the dynamics. This condition is found in the Sun's corona. If  $\beta \gg 1$ , the gas pressure is much larger than the magnetic pressure, so the plasma dominates and is able to push, stretch, and twist the magnetic field lines. This condition is found in the Sun's photosphere, where the dense, turbulent plasma drives the evolution of the magnetic field.

(c) *Briefly explain why measurements of the photospheric magnetic field — even though the flare happens much higher up — still give scientists useful information about where and when a flare might occur. Your answer should be no longer than a short paragraph.*

Although solar flares occur in the corona, the footpoints of the coronal magnetic loops are anchored in the photosphere. Motions of the dense plasma at the solar surface twist and stress these anchored magnetic field lines, storing magnetic energy in the corona. Therefore, measuring the photospheric magnetic field allows scientists to identify regions where magnetic energy is building up and estimate where and when a flare is likely to occur.

### 1.4 Task 4 - Reading the Sun's Magnetic "Warning Signs"

Scientists use data from NASA's Solar Dynamics Observatory (SDO) to forecast solar flares. The spacecraft measures the magnetic field at the Sun's surface in detail, and from these measurements a set of numerical parameters is calculated for each active region (a sunspot group). Machine-learning models are then trained on these parameters to predict whether a flare is likely.

Four of the most important parameters are:

Table 1: Summary of SHARP parameters used for solar flare prediction.

| Parameter | Description                                                                                                                                                                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R.VALUE   | A measure of how much magnetic flux of opposite polarity is packed tightly together along a boundary (called the polarity inversion line, or PIL). Think of it as measuring how sharply the magnetic field changes direction across a boundary. |
| TOTPOT    | A proxy for how much extra energy is stored in the magnetic field of the active region, above and beyond the minimum "relaxed" state. Think of it as measuring how "wound up" the region is.                                                    |
| TOTUSJH   | A measure of how twisted and helical the field lines are throughout the region. Think of it as measuring how much the field lines are spiralling, rather than running straight.                                                                 |
| TOTUSJZ   | A measure of the total electric current flowing vertically through the surface. Electric currents and twisted magnetic fields go hand in hand.                                                                                                  |

Suppose Active Region A has very large values of all four parameters. Active Region B has very small values of all four.

**(a) Using only the descriptions above (no equations required), explain in plain language what the magnetic field of Active Region A probably looks like, compared to Active Region B. Which region would you expect to be more likely to produce a large solar flare, and why?**

Active Region A is likely to have a much more complex and stressed magnetic field than Active Region B. A higher R.VALUE suggests that opposite magnetic polarities are packed closely together, creating a strong polarity inversion line. Higher values of TOTPOT, TOTUSJH, and TOTUSJZ indicate that the magnetic field stores more excess energy, is more twisted, and carries stronger electric currents. In contrast, Active Region B is likely to have a simpler, less twisted magnetic field with less stored energy. Therefore, Active Region A would be more likely to produce a large solar flare because it has more magnetic energy available to be released during magnetic reconnection.

**(b) Consider an analogy: a calm, flat lake versus a lake with large waves, strong**

*currents, and a whirlpool at the centre. Which lake stores more mechanical energy? How does this analogy map onto the difference between Active Region B and Active Region A?*

A lake with large waves, strong currents, and a whirlpool stores much more mechanical energy than a calm, flat lake because more energy is contained in its motion and turbulence. Similarly, Active Region A is like the turbulent lake: its magnetic field is more twisted, stressed, and carries stronger electric currents, meaning it stores much more magnetic energy. Active Region B is like the calm lake, with a simpler and more relaxed magnetic field that stores much less energy and is therefore less likely to produce a large solar flare.

*(c) These parameters are measured at the solar surface, but the flare happens in the corona above it. Explain in one or two sentences why a surface measurement can still act as a reliable warning sign for activity in the corona.*

Although solar flares occur in the corona, the magnetic loops where they originate are anchored at the photosphere. Motions at the solar surface twist and stress these anchored magnetic field lines, so measurements of the photospheric magnetic field reveal how much magnetic energy is being stored in the corona and can therefore indicate the likelihood of a flare.

*(d) Machine-learning models that use these parameters achieve better-than-chance forecasting of large solar flares, but they are not perfectly accurate. Suggest one physical reason why predicting exactly when a flare will occur is genuinely difficult, even with excellent measurements.*

Although the photospheric magnetic field reveals how much magnetic energy has been stored, the exact onset of a flare depends on when the coronal magnetic field becomes unstable and magnetic reconnection is triggered. These processes are highly complex and can be influenced by small changes in the magnetic field, making the exact timing of a flare difficult to predict.

## 2 Computational Module: The M5 Flare Problem

### 2.1 Task 1 - Exploratory Data Analysis

#### Modeling Decision

Summarize dataset shape, dtypes, missing values, duplicates, target distribution, feature distributions, and correlations. Conclude with which features look useful.

The training dataset contains 2,00,842 rows and 13 columns, with no missing values and no duplicate rows at the row level. *Inspecting every unique value in the target column*, however, reveals three distinct values rather than two: N (1,23,840), P (11,194), and **padding** (65,808, roughly 33% of the file). Every **padding** row was found to contain all-zero magnetic field measurements and a differently formatted timestamp compared to genuine observations, indicating these are synthetic filler rows rather than a valid third class. These rows were dropped entirely, as were NOAA\_AR and HARPNUM, since both are catalogue identification numbers rather than physical measurements and risk the model memorizing specific active regions instead of learning generalizable physics.

```

1      train_clean = train[train['Flare'] != 'padding'].copy()
2      train_clean = train_clean.drop(columns=['NOAA_AR', 'HARPNUM',
3      ↪      'DATE__OBS'])
4      train_clean['Flare'] = train_clean['Flare'].map({'N': 0, 'P': 1})

```

After cleaning, 135,034 rows remain, with a class ratio of 11.06:1 (N:P), matching the assignment's stated imbalance exactly (Figure 2).

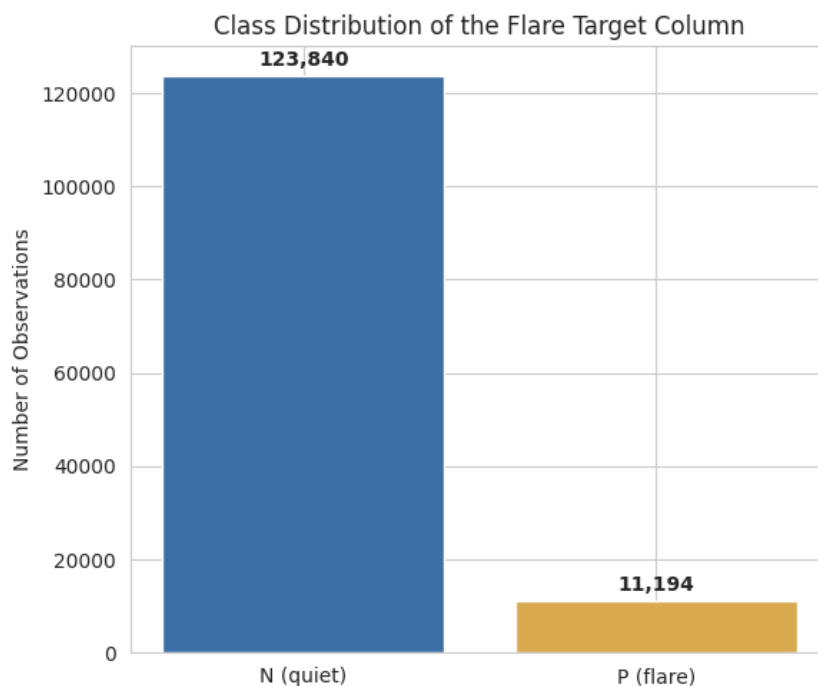


Figure 2: Class distribution of the Flare target column.

Feature distributions split by class (Figure 3) show clear separation for TOTUSJH, TOTPOT, R\_VALUE, ABSNJZH, SAVNCP, and USFLUX, and almost none for MEANALP. This is confirmed numerically: correlation with the target ranks TOTUSJH (0.292), TOTPOT (0.279), and ABSNJZH (0.260) highest, while MEANALP is essentially uncorrelated ( $-0.014$ ).

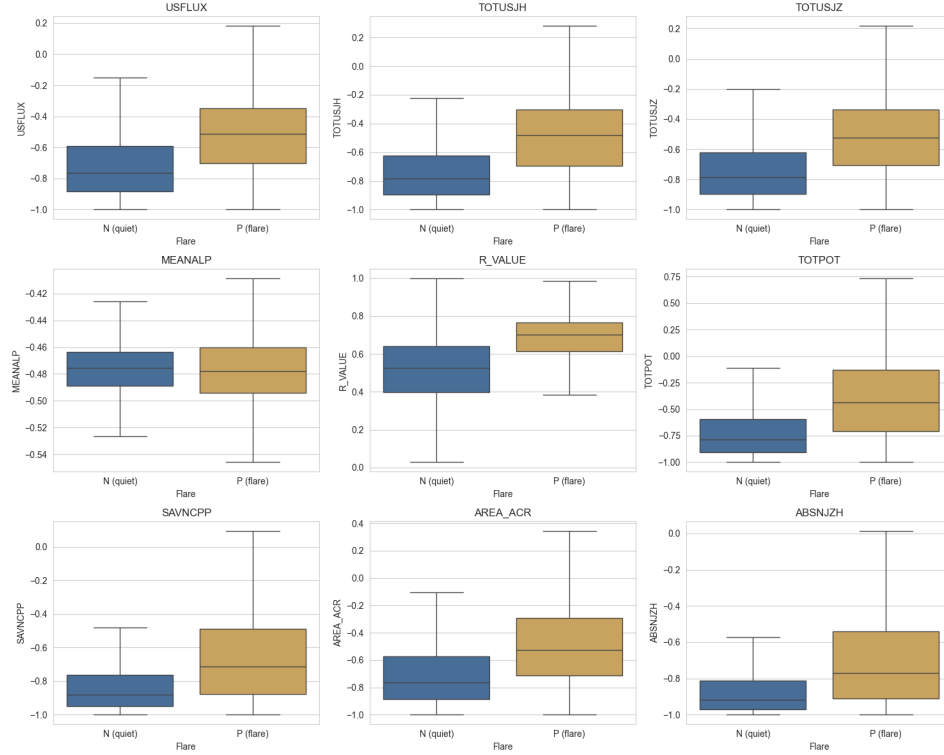


Figure 3: Feature distributions split by class (N = quiet, P = flare).

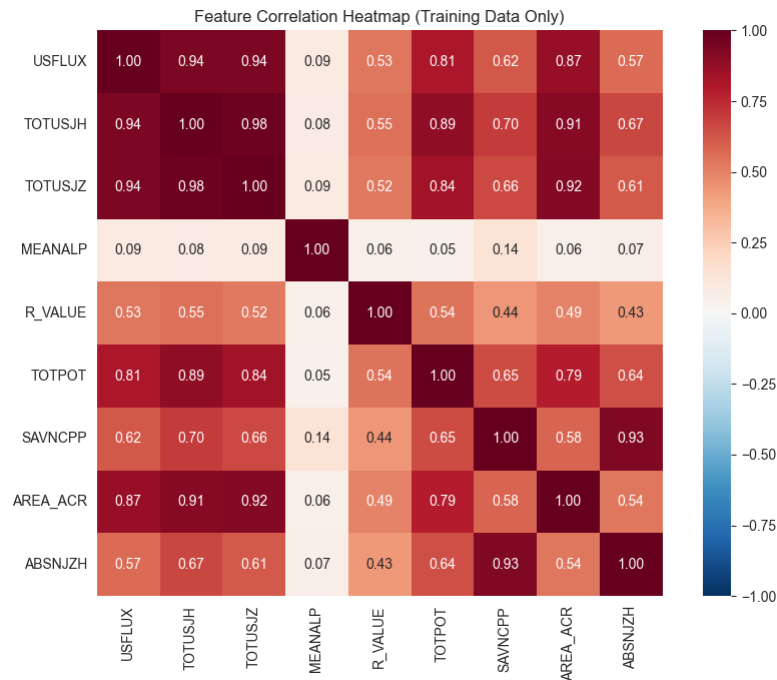


Figure 4: Feature-to-feature correlation heatmap (training data only).

**Conclusion:** TOTUSJH, TOTPOT, and ABSNJZH are the most useful features, consistent with the theoretical link between stored magnetic energy, current helicity, and flare likelihood. MEANALP carries almost no linear signal on its own - likely because it is a spatial average that cancels opposite-signed twist contributions – and Figure 4 shows substantial multicollinearity (correlations above 0.8) among several of the strongest features, which is not problematic for the tree-based models used in Task 4.

## 2.2 Task 2 - Handling Class Imbalance

### Modeling Decision

Baseline tree-based model (no resampling), then at least two imbalance-handling strategies (e.g. class weighting, SMOTE), each with rationale, trade-offs, and comparison to baseline.

A baseline Decision Tree, trained with no resampling and no class weighting, achieved 98.36% accuracy on a held-out validation split. *This figure alone is misleading:* a trivial classifier that always predicts “quiet,” never once identifying a real flare, already achieves 91.7% accuracy purely from the 11:1 class imbalance. Precision, recall, F1, and PR-AUC for the flare class are therefore used throughout instead of accuracy.

```
1 from sklearn.dummy import DummyClassifier
2 dummy = DummyClassifier(strategy='most_frequent').fit(X_train,
3               ↪ y_train)
4 print(dummy.score(X_val, y_val)) # 0.917 -- "free" accuracy from
5               ↪ imbalance
```

Two strategies were implemented and compared against this baseline:

1. **Class weighting** (`class_weight='balanced'`): reweights the loss function so misclassifying a rare flare costs more than misclassifying a quiet observation, without altering the dataset. *Advantage:* no synthetic data, no added training cost. *Disadvantage:* only reweights errors, without adding new information about the minority class's distribution.
2. **SMOTE**: generates synthetic flare examples by interpolating between real flare observations and their nearest flare neighbours in feature space. *Advantage:* directly increases minority-class representation with plausible new examples. *Disadvantage:* can generate unrealistic points where minority data is noisy, and adds training cost. SMOTE was fit exclusively on the training split to avoid leakage.

```
1 from imblearn.over_sampling import SMOTE
2 X_train_smote, y_train_smote =
3   ↪ SMOTE(random_state=42).fit_resample(X_train, y_train)
```

Table 2: Imbalance-handling strategy comparison (validation set).

| Strategy               | Precision | Recall | F1    | ROC-AUC | PR-AUC |
|------------------------|-----------|--------|-------|---------|--------|
| Baseline (no handling) | 0.910     | 0.889  | 0.900 | 0.941   | 0.819  |
| Class Weighting        | 0.922     | 0.895  | 0.908 | 0.944   | 0.834  |
| SMOTE                  | 0.880     | 0.927  | 0.903 | 0.958   | 0.822  |

Both strategies improved on the baseline. Class weighting gave a balanced, low-cost improvement across all metrics; SMOTE traded some precision for higher recall and ROC-AUC – arguably better aligned with the assignment’s stated cost asymmetry (missing a flare is costlier than a false alarm), though this trade-off was revisited with stronger models in Task 4.

### 2.3 Task 3 - Feature Engineering

#### Modeling Decision

At least five physically motivated engineered features, with reasoning and whether each improved performance.

Five features were engineered, each motivated by a physical mechanism from the Theoretical Module rather than an arbitrary transformation. Since all SHARP features are pre-scaled to  $[-1, 1]$ , engineered ratios are approximate proxies for physical quantities; any shift or clipping constant was fitted on the training set only.

Table 3: Engineered features and their physical motivation.

| Feature                  | Definition                               | Physical Motivation                                                                                                           |
|--------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| ENERGY_TWIST_INTERACTION | $\text{TOTPOT} \times \text{TOTUSJH}$    | Captures the interaction between stored magnetic energy and magnetic twist, both of which are associated with flare activity. |
| TWIST_AREA_PRODUCT       | $\text{MEANALP} \times \text{AREA\_ACR}$ | Preserves information on total magnetic twist that may be diminished by spatial averaging.                                    |
| PIL_ENERGY_CONCENTRATION | $\text{R\_VALUE} \times \text{TOTPOT}$   | Emphasizes magnetic energy concentrated near the polarity inversion line, a key region for flare initiation.                  |
| CURRENT_DENSITY_PROXY    | $\text{TOTUSJZ} / \text{AREA\_ACR}$      | Approximates current density by normalizing the total unsigned vertical current with active-region area.                      |
| HELICITY_PER_FLUX        | $\text{ABSNJZH} / \text{USFLUX}$         | Measures helicity normalized by the total magnetic flux, enabling comparisons across active regions of different sizes.       |

```

1      df['ENERGY_TWIST_INTERACTION'] = df['TOTPOT'] * df['TOTUSJH']
2      df['TWIST_AREA_PRODUCT']       = df['MEANALP'] * df['AREA_ACR']
3      df['PIL_ENERGY_CONCENTRATION'] = df['R_VALUE'] * df['TOTPOT']
4      df['CURRENT_DENSITY_PROXY']    = df['TOTUSJZ'] / (df['AREA_ACR'] -
5      ↪ area_shift + eps)
      df['HELICITY_PER_FLUX']         = df['ABSNJZH'] / (df['USFLUX'] -
      ↪ usflux_shift + eps)

```

Two ratio features (`CURRENT_DENSITY_PROXY`, `HELICITY_PER_FLUX`) produced extreme outliers where the denominator approached zero and were clipped to the 1st–99th percentile of the training set before use. Adding all five features to a class-weighted Decision Tree improved validation performance modestly but consistently:

Table 4: Impact of engineered features on validation performance.

| Feature Set             | Precision | Recall | F1    | ROC-AUC | PR-AUC |
|-------------------------|-----------|--------|-------|---------|--------|
| Original 9 features     | 0.922     | 0.895  | 0.908 | 0.944   | 0.834  |
| Original + 5 engineered | 0.920     | 0.904  | 0.912 | 0.948   | 0.839  |

Feature importance analysis showed `TWIST_AREA_PRODUCT` ranked 4th among all 14 features – notably higher than raw `MEANALP` (7th) – directly confirming the area-scaling hypothesis. `HELICITY_PER_FLUX` also contributed meaningfully (8th), while the remaining three engineered features added comparatively little, likely because they were built from features already highly intercorrelated with existing strong predictors (Figure 4).

## Task 4 – Modeling and Evaluation

### Modeling Decision

Decision Tree, Random Forest, AdaBoost, Gradient Boosting, XGBoost – hyperparameter choices, evaluation metrics for imbalance, ROC comparison, final model justification.

All five algorithms were trained on the final 14-feature set, tuned via randomized search with stratified cross-validation, scored on average precision (PR-AUC) rather than accuracy.

Table 5: Best hyperparameters found for each model.

| Model             | Best configuration                                                                    |
|-------------------|---------------------------------------------------------------------------------------|
| Decision Tree     | <code>max_depth=None, min_samples_leaf=10</code>                                      |
| Random Forest     | <code>n_estimators=150, max_depth=None, min_samples_leaf=1</code>                     |
| AdaBoost          | <code>base tree max_depth=8, n_estimators=150, learning_rate=0.3</code>               |
| Gradient Boosting | <code>n_estimators=150, max_depth=5, learning_rate=0.1</code>                         |
| XGBoost           | <code>n_estimators=200, max_depth=8, learning_rate=0.2, scale_pos_weight=11.06</code> |

*AdaBoost required particular care:* shallow decision stumps (depth 1–3), the conventional choice, produced near-zero recall, and applying `class_weight='balanced'` to the base estimator alongside AdaBoost’s own sample reweighting actively degraded performance. The fix was to supply balanced sample weights directly via `sample_weight` at fit time and increase base-tree depth to 8.

```

1      sample_weights = compute_sample_weight('balanced', y_train)
2      base = DecisionTreeClassifier(max_depth=8, random_state=42)
3      ada = AdaBoostClassifier(estimator=base, n_estimators=150,
4      ↪ learning_rate=0.3)
      ada.fit(X_train_final, y_train, sample_weight=sample_weights)

```

Table 6: Model comparison on the validation set.

| Model             | Precision | Recall | F1    | ROC-AUC | PR-AUC |
|-------------------|-----------|--------|-------|---------|--------|
| Decision Tree     | 0.710     | 0.941  | 0.809 | 0.968   | 0.899  |
| Random Forest     | 0.991     | 0.931  | 0.960 | 0.999   | 0.996  |
| AdaBoost          | 0.719     | 0.957  | 0.821 | 0.994   | 0.959  |
| Gradient Boosting | 0.612     | 0.957  | 0.747 | 0.990   | 0.918  |
| XGBoost           | 0.959     | 0.975  | 0.967 | 0.999   | 0.993  |

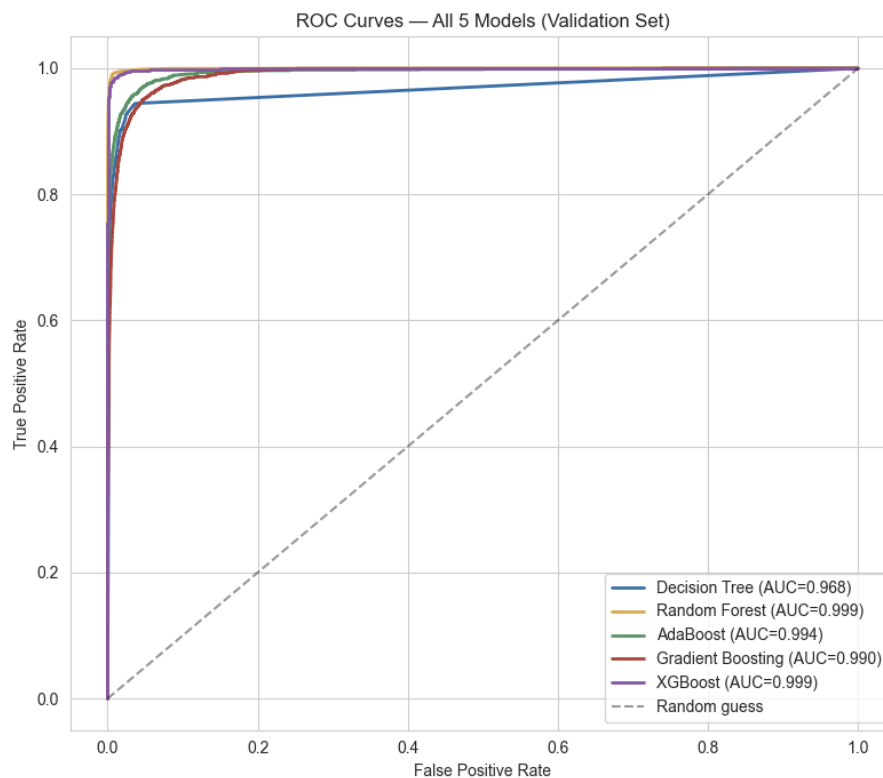


Figure 5: ROC curves for all tuned models.

**Final model selection: XGBoost.** Random Forest and XGBoost clearly outperform the other three models on every metric. Between the two, XGBoost was selected because it catches meaningfully more real flares than Random Forest (97.5% vs. 93.1% recall) at only a small cost in precision (95.9% vs. 99.1%) – the more appropriate trade-off given the assignment’s statement that missing a flare is costlier than a false alarm. This decision is deliberately not based on accuracy nor purely on PR-AUC, but on the recall/precision trade-off aligned with the stated cost asymmetry.

### 2.3.1 Final Evaluation on the Held-Out Test Set

The test set was opened only after the model above was finalized, using the identical cleaning and feature-engineering pipeline with constants fitted on the training set only.

Table 7: XGBoost performance on the held-out test set.

| Metric        | Value |
|---------------|-------|
| Precision (P) | 0.024 |
| Recall (P)    | 0.026 |
| F1 (P)        | 0.025 |
| ROC-AUC       | 0.504 |
| PR-AUC        | 0.020 |

Of 231 real flares in the test set (imbalance ratio 63.95:1, far worse than training’s 11:1), only 6 were correctly identified, with 245 false alarms raised on quiet regions. An ROC-AUC of 0.504 is indistinguishable from random guessing – serious enough to investigate rather than accept at face value. Two checks confirmed this is a genuine distribution shift rather than a pipeline error: the train/test split is *temporal* (training spans January 2023 to February 23, 2025; testing spans February 23 to April 27, 2025, a genuine future period), and test-set active regions are systematically weaker than training-set regions even among confirmed flares (training flares averaged a scaled TOTPOT of  $-0.372$ , maximum 0.975; test flares averaged  $-0.771$ , maximum only  $-0.714$ ). All five models collapsed similarly on the test set (ROC-AUC 0.50–0.67), confirming this is a property of the data split, not one algorithm’s quirk.

## 2.4 Task 5 - Scientific Conclusion

Discuss learned patterns, successes/misses, physical explanations, limitations, and answer the reflection question on whether missing most M5 flares means the model is bad.

The model learned a physically sound pattern from the training distribution, relying most heavily on TOTUSJH, TOTPOT, R\_VALUE, and TWIST\_AREA\_PRODUCT – all consistent with the reconnection-driven flare mechanism described in the Theoretical Module. Of the few test-set flares correctly identified, the clearest separating feature from missed flares was R\_VALUE (0.670 vs. 0.492 on average), suggesting that polarity-inversion-line concentration – a more locally-defined, ratio-like

quantity – shifted less between training and test periods than absolute-magnitude quantities such as TOTPOT.

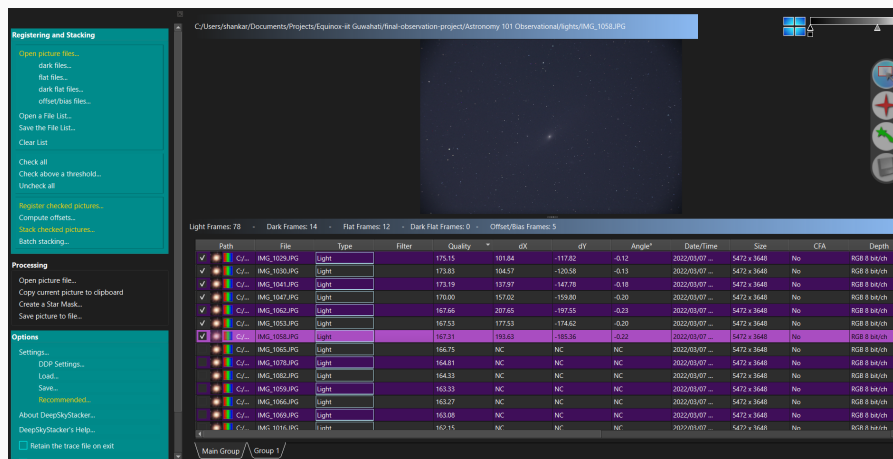
The most plausible physical explanation for the widespread misses is that the test window falls in a different phase of the roughly 11-year solar activity cycle than most of the training window (Solar Cycle 25 was expected to peak around 2024), such that active regions in the test period are systematically smaller and less energetic even when they do produce a flare. None of the five algorithms tested include any mechanism for adapting to a shifting feature distribution over time, and the training window (roughly 2.3 years) is likely too short to have captured a full cycle’s range of behaviour.

*Does missing most flares mean the model is bad?* Not necessarily. Every stage of the workflow was executed soundly: the target column was properly cleaned, class imbalance was addressed with justified strategies, features were engineered from physical reasoning and validated via ablation testing, five algorithms were tuned via cross-validation on training data alone, and the test set was kept sealed until final evaluation. The model performed excellently on validation data drawn from the same distribution as training ( $F1 = 0.967$ ,  $PR-AUC = 0.993$ ). The test-set failure is better understood as a genuine scientific finding about the limits of the available training data than as a methodological error: a model can only learn relationships present in the data it was shown, and if the underlying physical regime shifts between training and deployment, no amount of tuning can compensate for information the training data never contained. This itself carries operational value, indicating that solar flare prediction models trained on a limited window of the solar cycle should be periodically retrained rather than deployed indefinitely on a fixed training set.

## 3 Observational Module: The Galactic Neighbour

### 3.1 Task 1 – Calibration & Integration (DeepSkyStacker)

#### Workflow Screenshot

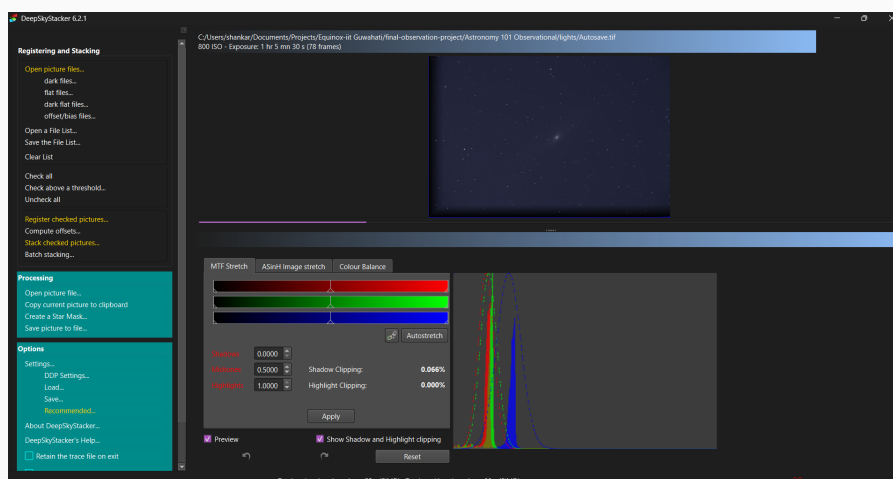


The screenshot shows the DeepSkyStacker interface with a list of 98 light frames. The table below represents the data shown in the frame list.

| Path                                                                                                                                     | File          | Type  | Filter | Quality | dX     | dY      | Angle | Date/Time      | Size        | CFA | Depth        |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|--------|---------|--------|---------|-------|----------------|-------------|-----|--------------|
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1029.JPG | 101s_1029.JPG | Light |        | 175.15  | 101.84 | -117.82 | -0.12 | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1041.JPG | 101s_1041.JPG | Light |        | 175.83  | 104.57 | -120.58 | -0.13 | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1042.JPG | 101s_1042.JPG | Light |        | 174.19  | 137.97 | -141.78 | -0.18 | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1043.JPG | 101s_1043.JPG | Light |        | 170.00  | 159.02 | -159.80 | -0.20 | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1044.JPG | 101s_1044.JPG | Light |        | 187.66  | 207.65 | -197.55 | -0.23 | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1053.JPG | 101s_1053.JPG | Light |        | 187.53  | 177.53 | -174.62 | -0.20 | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1058.JPG | 101s_1058.JPG | Light |        | 187.31  | 193.63 | -185.36 | -0.22 | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1059.JPG | 101s_1059.JPG | Light |        | 186.75  | NC     | NC      | NC    | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1078.JPG | 101s_1078.JPG | Light |        | 186.81  | NC     | NC      | NC    | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1080.JPG | 101s_1080.JPG | Light |        | 184.33  | NC     | NC      | NC    | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1089.JPG | 101s_1089.JPG | Light |        | 183.33  | NC     | NC      | NC    | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1096.JPG | 101s_1096.JPG | Light |        | 183.27  | NC     | NC      | NC    | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1106.JPG | 101s_1106.JPG | Light |        | 183.08  | NC     | NC      | NC    | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |
| C:\Users\shankar\Documents\Projects\Equinox-101\Gawabhai\final-observation-project\Astronomy\101 Observational\lights\101s\101s_1116.JPG | 101s_1116.JPG | Light |        | 182.15  | NC     | NC      | NC    | 2022/03/07 ... | 5472 x 3648 | No  | RGB 8 bit/ch |

Caption: Frame list showing registration scores (Quality, FWHM, #Stars,  $dX/dY/Angle$ ) for all 98 Light frames. Frames were filtered using DeepSkyStacker's "Check above a threshold" function, set to  $Quality \geq 167$ . This threshold was chosen because frames below it corresponded closely with those that failed automatic star-pattern registration (marked "NC" for  $dX/dY/Angle$ ), which also showed systematically lower star counts and higher FWHM (star bloat) than successfully registered frames – consistent with focus drift, tracking error, or thin cloud passing during those exposures. This excluded the weakest frames while retaining the majority of well-registered frames for stacking.

#### Workflow Screenshot

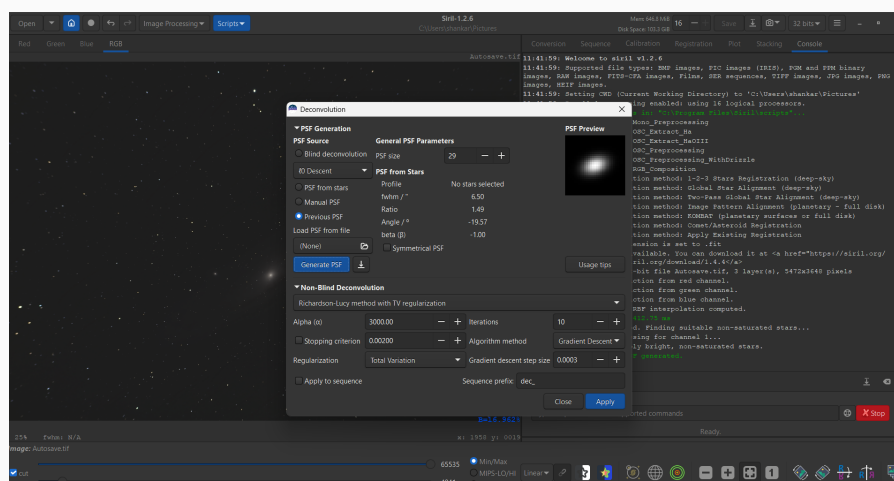


Caption: Final stacked master image produced by DeepSkyStacker, generated from the checked (high-quality) Light frames after calibration against the corresponding Dark, Flat, and Bias/Offset frames. Kappa-Sigma clipping was used during stacking to reject

outlier pixels (e.g. satellite trails, cosmic ray hits) that appeared in only a subset of frames.

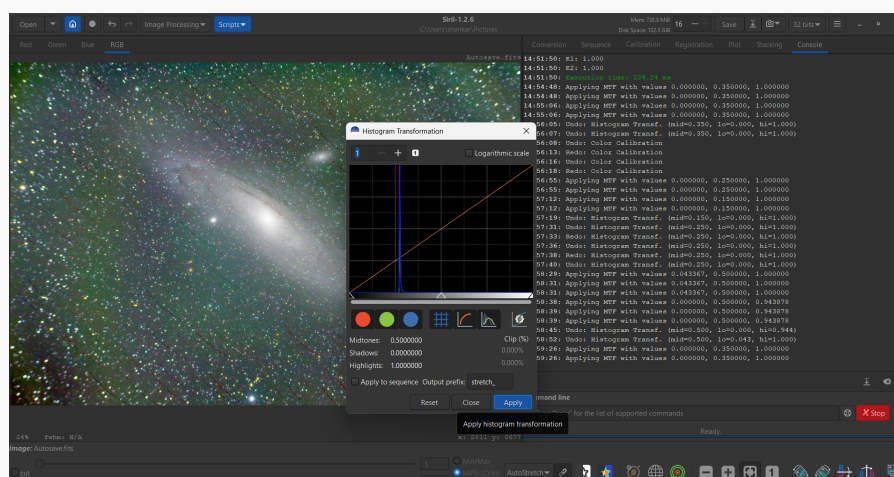
### 3.2 Task 2 – Signal Extraction & Color Calibration (Siril)

#### Workflow Screenshot



Caption: Deconvolution window in Siril, applied after background extraction (RBF model) to the linear master stack. A point spread function (PSF) was estimated from a bright star in the field, and deconvolution was applied to sharpen stellar profiles and recover finer structural detail before stretching.

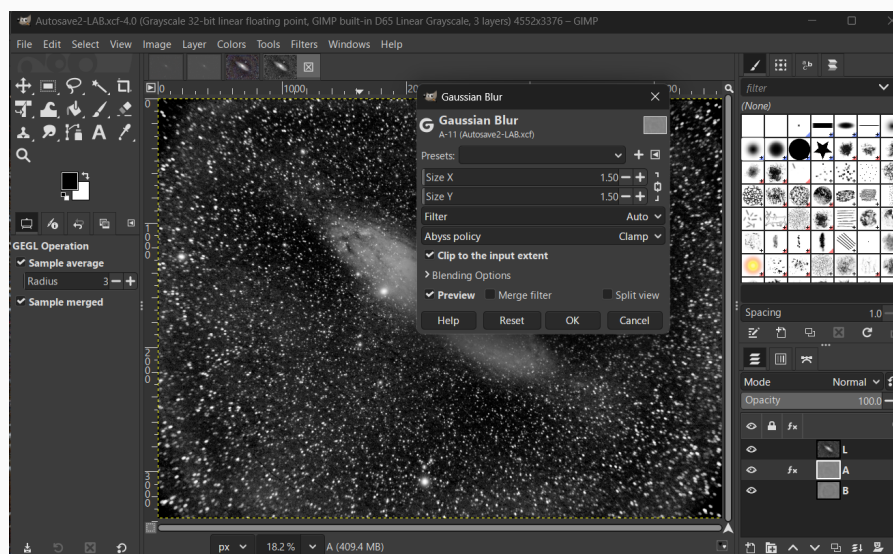
#### Workflow Screenshot



Caption: Image immediately after applying a non-linear histogram stretch (Generalized Hyperbolic Transformation) in Siril, with the corresponding stretch curve shown. This step converts the linear-intensity stacked data – where faint galaxy structure is compressed near black – into a viewable image revealing the spiral disk, dust lane, and surrounding star field. The stretched image was exported as a 32-bit TIFF for further processing.

### 3.3 Task 3 – Noise Mitigation & Final Polish (GIMP)

#### Workflow Screenshot



*Caption: The stretched image decomposed into LAB color space in GIMP, showing the L (Lightness), A (green-red), and B (blue-yellow) layers, with the Gaussian Blur dialog open on the A channel. Since chrominance noise predominantly resides in the A/B channels while structural detail resides in L, blurring only A and B targets color noise without softening stars or galaxy detail.*

#### Modeling Decision

Due to GIMP rendering-precision constraints encountered when previewing low-contrast A/B channel data (the decomposed document defaulted to a linear light encoding, which visually compressed the effect of the blur to the point of being indistinguishable on screen), the recompose step was found to be unreliable in this GIMP session. Final chrominance noise reduction was instead completed using GIMP's built-in **Filters** → **Enhance** → **Noise Reduction** filter, applied directly to the correctly-composed color image. This filter targets the same underlying noise characteristics (smoothing low-amplitude chrominance variation while preserving edges/detail) and was verified visually via GIMP's Undo History, comparing the image immediately before and after the filter at 100% zoom on the same background region.

A modest saturation boost (Colors → Hue-Saturation) was applied afterward to compensate for the slight desaturation typically introduced by chrominance noise reduction.

### 3.4 Final Visual Product



Figure 6: Final processed image of the Andromeda Galaxy (M31), showing the spiral disk, prominent dust lane, diffuse outer halo, and the companion galaxy M32 (upper right).